[CSE110 Spring 2024 Lab Exam - 0X Tentative Solutions and Rubrics]

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A]

SOLUTION

```
import java.util.Scanner;

public class DataUsageTracker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter N: ");
        int N = sc.nextInt();

        int dataScore = 0;

        for (int i = 1; i <= N; i++) {
            System.out.print("Enter data for day " + i + ": ");
            int data = sc.nextInt();

            if (data < 50 || data > 5000) {
                System.out.println("Invalid data amount, " + data + ".");
                i--;
                continue;
            }

            if (data % 4 == 0 || data % 10 == 5) {
                dataScore += data;
            } else {
                dataScore -= data;
            }
        }

        int scoreCopy = dataScore;
        if (scoreCopy < 0) scoreCopy *= -1;

        int evenCount = 0;
        while (scoreCopy > 0) {
            int digit = scoreCopy % 10;
            if (digit % 2 == 0) evenCount++;
            scoreCopy /= 10;
        }

        System.out.println("Final Data Score: " + dataScore);
        System.out.println("Number of even digits: " + evenCount);
    }
}
```

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RUBRIC

SL	Points To Meet	Marks [10]
1	Scanner used properly, N input taken	1.5
2	Loop runs N times with re-prompt	1
3	Proper Range check (50–5000)	1.5
4	Score updated using rules	2.5
5	Even-digit counting + negative handling	2.5
6	Output displayed correctly	1

[SET-B]

SOLUTION

```
import java.util.Scanner;

public class WaterConsumptionTracker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter N: ");
        int N = sc.nextInt();

        int hydrationScore = 0;

        for (int i = 1; i <= N; i++) {
            System.out.print("Enter water for day " + i + ": ");
            int water = sc.nextInt();

            if (water < 100 || water > 10000) {
                System.out.println("Invalid water amount, " + water + ".");
                i--;  // re-prompt this day
                continue;
            }

            if (water % 5 == 0 || water % 10 == 2) {
                hydrationScore += water;
            } else {
                hydrationScore -= water;
            }
        }

        int scoreCopy = hydrationScore;
        if (scoreCopy < 0) scoreCopy *= -1;

        int oddCount = 0;
        while (scoreCopy > 0) {
            int digit = scoreCopy % 10;
            if (digit % 2 != 0) oddCount++;
            scoreCopy /= 10;
        }

        System.out.println("Final Hydration Score: " + hydrationScore);
        System.out.println("Number of odd digits: " + oddCount);
    }
}
```

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RUBRIC

SL	Points To Meet	Marks [10]
1	Scanner creation & input N taken	1.5
2	Loop runs N times, reprompt on invalid input	1
3	Range check 100–10,000 ml	1.5
4	Score updated based on divisible by 5 or ends with 2 rule	2.5
5	Odd digit counting with negative handling	2.5
6	Final output formatting correct	1